

Limits – 4

Upper and lower limits, subsequences, Stolz

Homework

1 сентября 2026 года

Theoretical minimum. Upper and lower limits and their properties; the set of partial limits and its closedness; the Cauchy criterion; the Bolzano–Weierstrass theorem and the existence of a monotone subsequence; the Stolz–Cesàro theorem; Toeplitz’s lemma and regular summation methods; the formula $\limsup \sqrt[n]{a_n}$ in Cauchy’s root test.

1. Prove that $\liminf \frac{a_{n+1}}{a_n} \leq \liminf \sqrt[n]{a_n} \leq \limsup \sqrt[n]{a_n} \leq \limsup \frac{a_{n+1}}{a_n}$ for positive a_n , and give an example in which both outer inequalities are strict.
2. Let $x_n > 0$ satisfy $x_{n+m} \leq x_n x_m$ for all n, m . Prove that $\sqrt[n]{x_n}$ has a limit equal to $\inf_n \sqrt[n]{x_n}$ (the multiplicative Fekete lemma), and compare with the additive version from the first sheet.
3. (Proposed problems, №265) Let $x_n > 0$, $x_n \rightarrow 0$ and $\sum x_n = \infty$. Prove that there is a subsequence (x_{n_k}) with $\sum x_{n_k} = \infty$ and $\sum x_{n_k}^2 < \infty$.
4. (Proposed problems) Suppose $\limsup_{n \rightarrow \infty} (1 - x_n) \ln n < \infty$. Prove that if a series $\sum a_n$ with positive terms converges, then so does $\sum a_n^{x_n}$.
5. Prove that every sequence contains a monotone subsequence and deduce the Bolzano–Weierstrass theorem; state and prove the analogue in $\mathbb{R}^d$.
6. Suppose the subsequences (x_{2n}) , (x_{2n+1}) and (x_{3n}) all converge. Prove that the whole sequence converges. Does the statement remain true if the subsequence (x_{3n}) is dropped?
7. Prove that $\limsup (x_n y_n) \leq \limsup x_n \cdot \limsup y_n$ for non-negative sequences, and describe all cases of equality when $\limsup x_n > 0$.

8. Let $a_n \rightarrow a$. Prove that $\frac{a_1 + 2a_2 + \cdots + na_n}{n^2} \rightarrow \frac{a}{2}$, and that $\sqrt[n]{a_1 a_2 \cdots a_n} \rightarrow a$ for positive a_n ; deduce both from Toeplitz's lemma.
9. Construct a sequence whose set of partial limits is exactly the set of rational numbers of $[0, 1]$, and prove that the set of partial limits can never be exactly the set of irrational numbers of that interval.
10. Let x_n be bounded with $2x_n \leq x_{n-1} + x_{n+1}$ for all $n \geq 2$ (discrete convexity). Prove that x_n converges, and determine the possible values of $\lim(x_{n+1} - x_n)$.